

INTERNATIONAL news

US to expand list of countries for EV credit eligibility

The US Department of the Treasury said that countries other than the US and its free trade agreement partners may be added to the Inflation Reduction Act as places of production eligible for US EV credit. The existing rules of the IRA suggest that for any EV maker to be eligible for the credit, the product needs to undergo final assembly in North America. It allows the imports of critical minerals from countries with free trade agreements to still qualify for incentives, but not imports of battery components. The law forbids tax credits for vehicles with any components or critical minerals sourced from a "foreign entity of concern," such as China or Russia.

Samsung plans to raise chip production in 2023

As per various reports, Samsung Electronics intends to raise semiconductor chip production capacity in 2023 at its largest semiconductor plant in South Korea amid forecasts of an economic slowdown. The company plans to expand its P3 factory in Pyeongtaek by adding a 30.4cm (12-inch) wafers capacity for dynamic RAM (DRAM) memory chips. Earlier, Samsung announced the development of its 16GB DRAM using the industry's first 12nm-class process technology.

Foxconn partners with Nvidia to build autonomous vehicles

Electronics contract manufacturer Hon Hai Technology Group (trading as Foxconn Technology Group) has partnered with graphics semiconductor chip manufacturer Nvidia to develop automated and autonomous vehicle platforms. Foxconn will be producing cars based on Nvidia's DRIVE Orin electronic control units (ECUs) and DRIVE Hyperion sensors. Nvidia's DRIVE Hyperion sensor is a modular development platform and reference architecture for designing autonomous vehicles. The automotive-grade NVIDIA DRIVE Orin system-on-chip (SoC) achieves up to 254 trillion operations per second.

Baidu and Pony.ai to test autonomous robotaxis without safety backup

Baidu Inc. and Toyota Motor Corp backed startup Pony.ai will soon be testing fully autonomous vehicles without safety operators as a backup in Beijing. The two companies are among the first to be granted licences to offer commercial robotaxi services in China's capital. The two companies said they would test ten driverless vehicles in a technology park developed by the Beijing government. Chinese regulations currently allow Level 4 autonomous driving capability, which does not require a human driver to control the vehicle, in robotaxis that operate within designated geofenced areas.

TSMC announces mass production of 3nm chips in Taiwan

Taiwan Semiconductor Manufacturing Company (TSMC) has announced mass production of its 3nm chips in Southern Taiwan. The company's chairman said it would continue to expand capacity on the island. The 3nm chips are considered to be the most advanced chips produced by TSMC, which will outperform the 5nm chips found in the latest Pro iPhone models. This announcement comes six months after Samsung achieved 3nm production.

Kyocera to double CAPEX and R&D expenditure over next three years

Japanese electronics maker Kyocera Corp intends to double

its total capital investment (CAPEX) and research and development expenditure to \$9.78 billion over the next three fiscal years through March 2026 to expand its manufacturing capabilities. As per reports, Kyocera doubled the amount spent during the past three years to March 2023 to fund the expansion of parts such as ceramics for chip-making equipment. It plans to spend up to \$6.728 billion on CAPEX, which is double the amount it has spent over the past three years.

Samaiden launches rooftop solar PV project at Sunway Nexis

Samaiden Group Berhad, a clean energy solution provider, announced that their wholly-owned subsidiary Samaiden Capital Management Sdn Bhd's first solar investment project at Sunway Nexis commenced operation in late December. Samaiden



will operate and maintain the rooftop solar PV system with an installed capacity of 531 kilowatt-peak (kWp) for a duration of 20 years. It believes that around 8,400 tonnes of carbon emissions can be reduced, owing to the generation of 12.5 million kilowatt hours (kWh) of energy over the 20-year period.

Taiwan chipmakers can turn 25% of R&D expense into tax credits

Taiwan has passed new rules that allow local chip firms to turn 25% of their annual research and development expenses into tax credits to retain the island's dominance in the semiconductor market. The new incentives are expected to be brought into effect sometime in 2023. Chip companies in Taiwan can also claim tax credits on 5% of the annual costs of buying new equipment for advanced process technologies, the ministry said.

Aries and Emdalo team up to develop RISC-V and FPGA modules

Germany based Aries Embedded has partnered with Emdalo Technologies to develop embedded modules based on Microchip's PolarFire RISC-V SoC and FPGA architecture. The PolarFire FPGA architecture, used by Aries, includes multiple embedded RISC-V processors for its M100PF and M100PFS system-on-modules (SoM). These are already used in a range of applications, including smart embedded vision, industrial automation, communications, and IoT. The M100PF SoM family implements the PolarFire FPGA that spans from 100K logic elements (LEs) to 300K LEs. It features 12.7G transceivers and needs up to 50% lower power than competing mid-range FPGAs.